

CMS Open Data H to Tau Tau Search: Final Analysis Note

Analysis my_analysis

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Abstract

This note reports a full rerun of the muon plus hadronic-tau Higgs-to-tau-tau analysis with the current local CMS Open Data and simulation inputs. The analysis follows the Run-1 H to tau tau strategy of a simultaneous binned fit in VBF, boosted, and zero-jet categories (CMS Collaboration 2014) and keeps two result streams: a cut-based visible-mass baseline and a single three-category D_{NN} classifier-score result. The visible-mass baseline gives `mu_hat` = 2.4740, observed 95% CLs `mu` < 5.9262, median expected limit 3.6919, and interval proxy `mu_hat` = 2.4740 +3.4522 -2.4740. The D_{NN} score result gives `mu_hat` = 1.6160, observed 95% CLs `mu` < 3.5769, median expected limit 1.8069, and interval proxy `mu_hat` = 1.6160 +1.9610 -1.6160. The results are interpreted as a single-channel public-data exercise, not as a replacement for the published CMS combination (CMS Collaboration 2018).

1 Introduction

The target process is Standard Model Higgs boson production followed by $H \rightarrow \tau\tau$, reconstructed in the $\mu\tau_h$ final state. This channel combines a clean isolated muon with a hadronic-tau candidate and is one of the classic ingredients of the CMS H to tau tau program (CMS Collaboration 2014). The parameter of interest is the signal-strength modifier

$$\mu = \frac{\sigma(pp \rightarrow H) \mathcal{B}(H \rightarrow \tau\tau)}{[\sigma(pp \rightarrow H) \mathcal{B}(H \rightarrow \tau\tau)]_{SM}}. \quad (1)$$

The reduced Open Data setting has less information than the published Run-1 analyses: embedded $Z \rightarrow \tau\tau$ samples, several rare backgrounds, and full object-calibration machinery are not available. The note therefore emphasizes the reproducible signal-extraction chain, the category-level validation, and the relative behavior of the visible-mass and D_{NN} observables.

2 Data And Simulation

The data input is the localized CMS 2012 TauPlusX Run2012B and Run2012C sample. The simulated inputs are ggH and VBF Higgs signal, DY+jets, top-pair, and exclusive W+jets samples. The normalization uses the Open Data record event counts and the luminosity reference stored in the local normalization artifact.

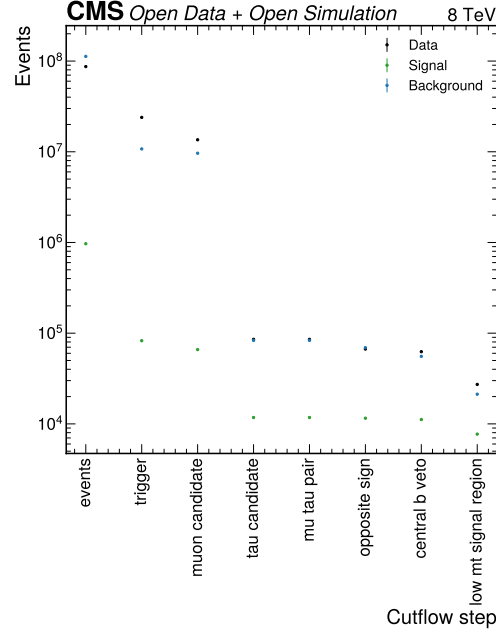


Figure 1: Input and selected-event cutflow. The figure shows the data and simulation counts through the main selection stages after the rerun on the current local inputs. It is the first check that the event processing consumed the updated ROOT files.

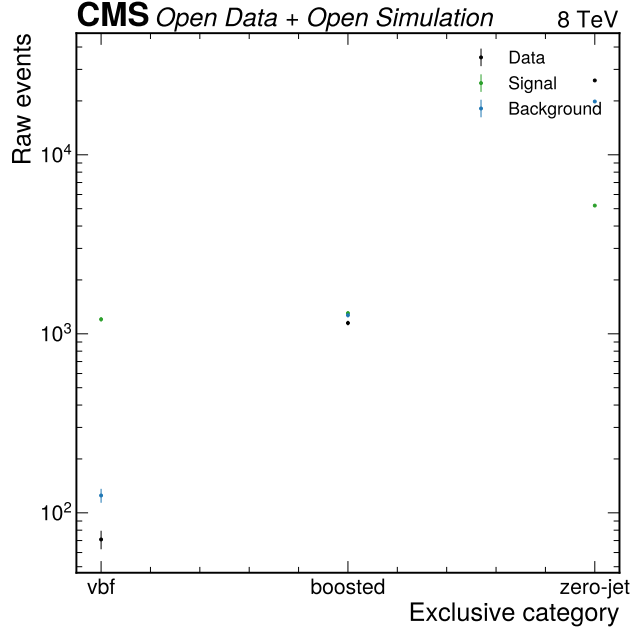


Figure 2: Final category yields. The figure summarizes the selected yields in the VBF, boosted, and zero-jet categories before the final binned statistical fit. It provides a compact check of the category split used by both retained observables.

3 Event Selection And Categories

Events are required to contain a selected muon, a selected hadronic-tau candidate, low transverse mass, and an opposite-sign signal-region pair. The categories are mutually exclusive. The VBF category requires two clean jets with a large dijet mass and pseudorapidity separation, the boosted category captures remaining events with a reconstructed recoil topology, and the zero-jet category contains the statistically dominant inclusive remainder.

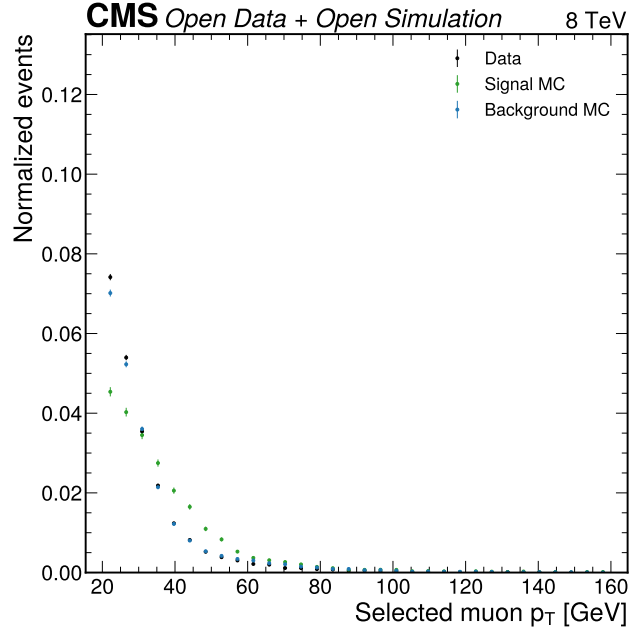


Figure 3: Selected muon transverse momentum. The comparison shows the selected muon p_T spectrum in data and simulation after the signal-region selection. It is used as an object-level validation input rather than a fitted observable.

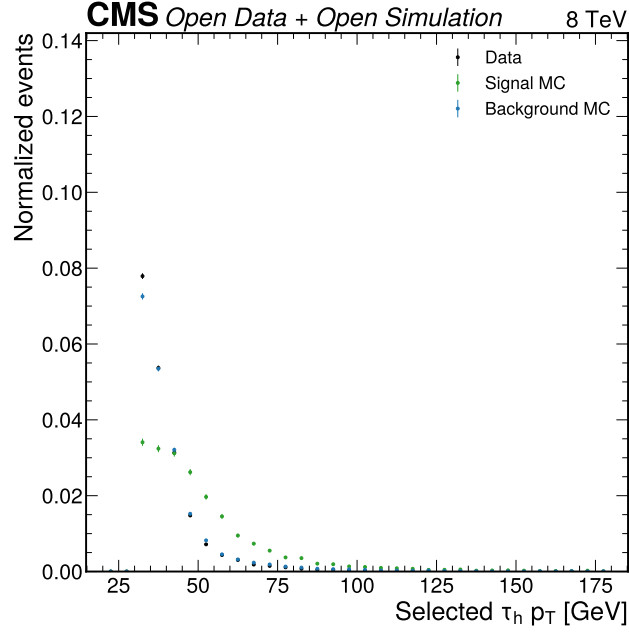


Figure 4: Selected hadronic-tau transverse momentum. The comparison shows the selected $\tau_h p_T$ spectrum in data and simulation. It checks the stability of the tau-object phase space used by the baseline and D_{NN} fits.

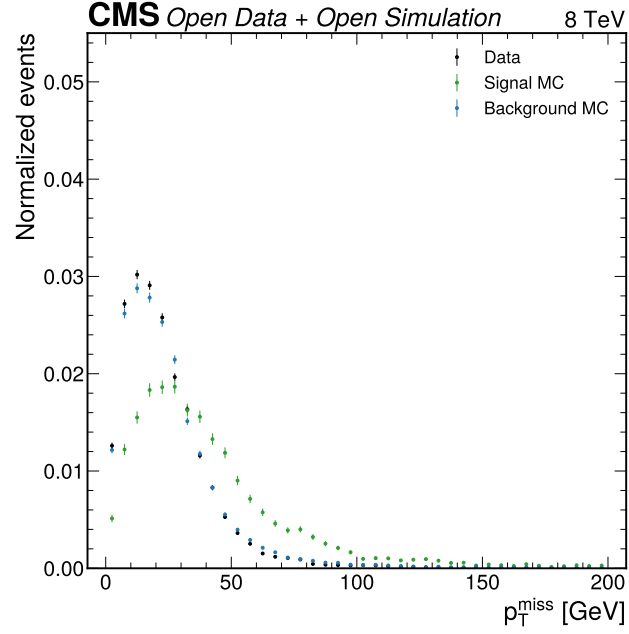


Figure 5: Missing transverse momentum. The comparison shows the selected missing transverse momentum distribution. This variable enters the classifier input set and is also a key validation distribution for W+jets and tau-pair kinematics.

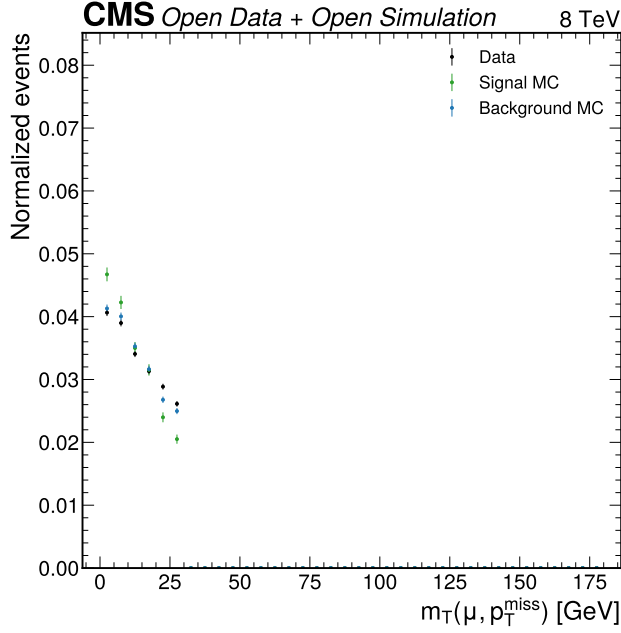


Figure 6: Transverse mass control distribution. The comparison shows the $m_T(\mu, p_T^{miss})$ distribution used to monitor the W+jets-rich high-mass region. The final W+jets scale is derived from the corresponding high- m_T control selection.

4 Observables

The baseline observable is the visible di-tau mass m_{vis} , the invariant mass of the reconstructed muon and hadronic-tau four-vectors. It is robust and cut-based, but broad because neutrinos from both tau decays are not reconstructed. The retained multivariate observable is D_{NN} , implemented as an XGBoost classifier score `mva_score_xgboost` in 20 uniform bins from 0 to 1. Its input set is:

`m_vis, mu_pt, mu_eta, tau_pt, tau_eta, delta_r_mu_tau, delta_phi_mu_tau, met_pt, met_significance, mt_mu_met, mt_tau_met, mt_tot, pt_tautau_proxy, m_coll, tau_id_iso_raw.`

The classifier is trained on MC signal and background labels with the nominal MC normalization weights and class balancing. In the current rerun the stored training metric is test AUC 0.8247 with train AUC 0.8524.

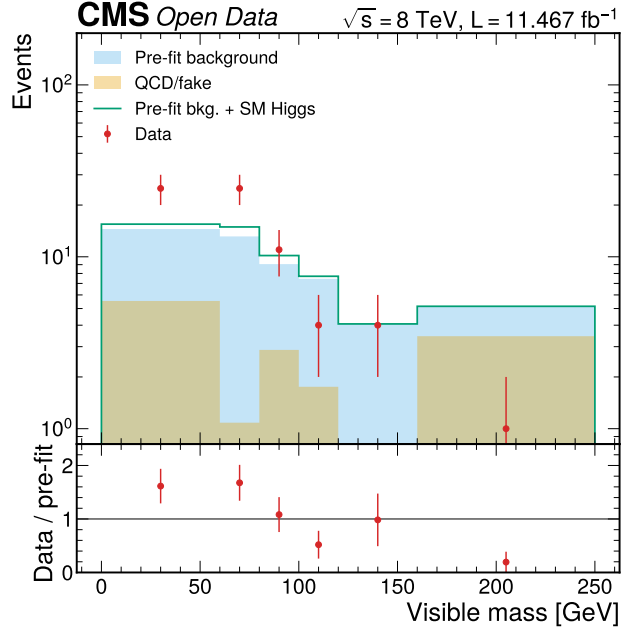


Figure 7: Baseline visible-mass pre-fit validation in the VBF category. The plot compares data with the nominal visible-mass background plus Standard Model Higgs prediction before profiling nuisance parameters in the signal fit. The lower panel is Data/pre-fit.

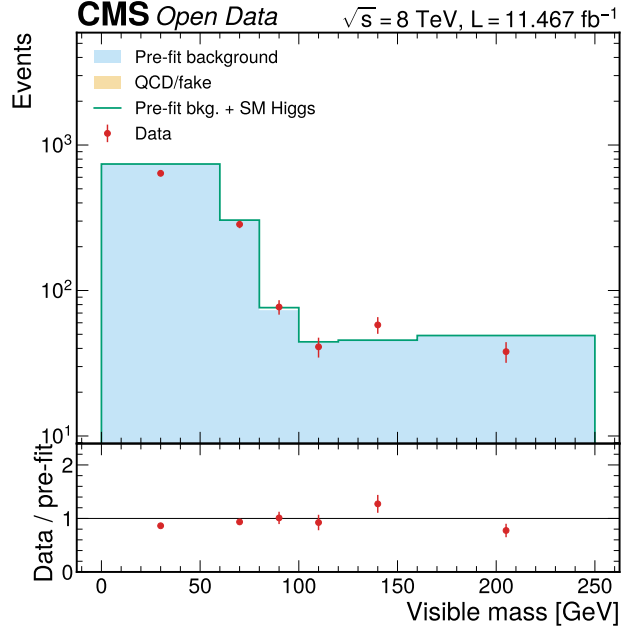


Figure 8: Baseline visible-mass pre-fit validation in the boosted category. The plot shows the boosted-category nominal visible-mass prediction before the post-fit profile update. This is a major contributor to the simultaneous visible-mass baseline fit.

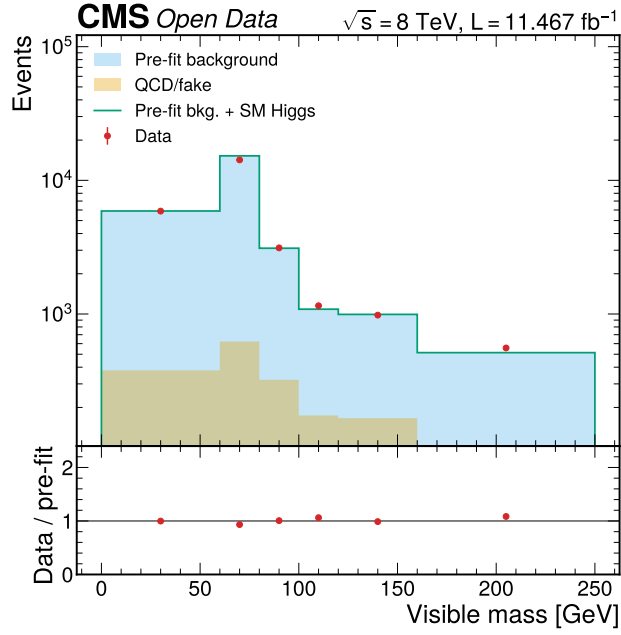


Figure 9: Baseline visible-mass pre-fit validation in the zero-jet category. The plot shows the highest-statistics baseline category before post-fit nuisance profiling. It anchors the background-rich part of the simultaneous fit.

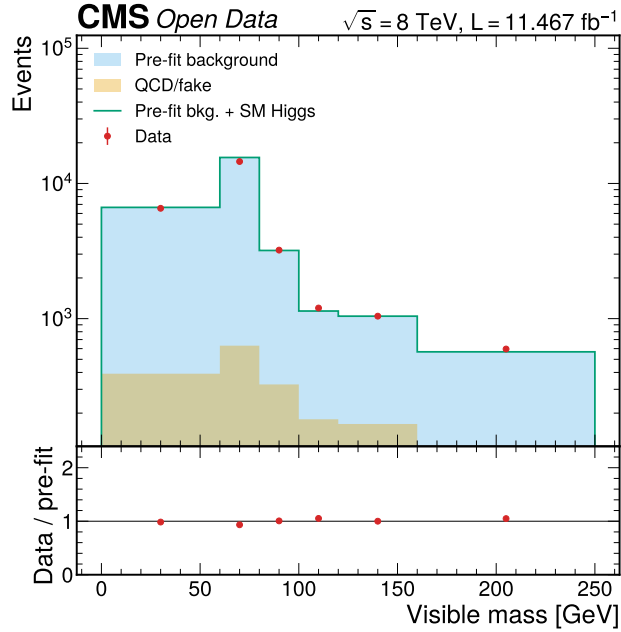


Figure 10: Baseline visible-mass combined pre-fit validation. The plot sums the VBF, boosted, and zero-jet nominal visible-mass predictions and compares them with the summed observed data before profiling. The lower panel is the combined Data/pre-fit ratio.

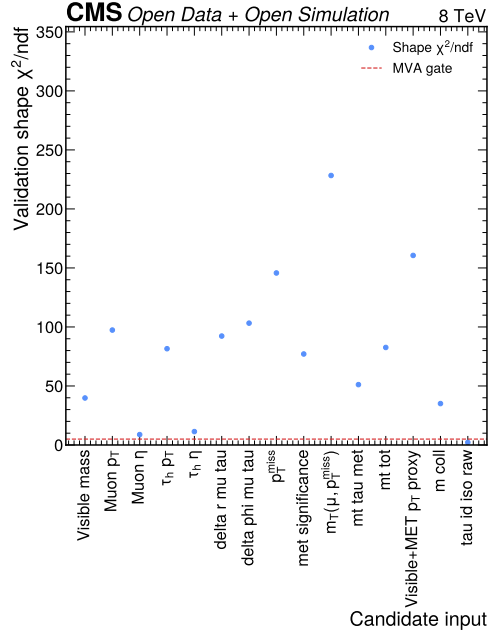


Figure 11: MVA input modelling summary. The figure records the data/background modelling quality of candidate classifier inputs. It documents the validation context for using a single retained D_{NN} tool.

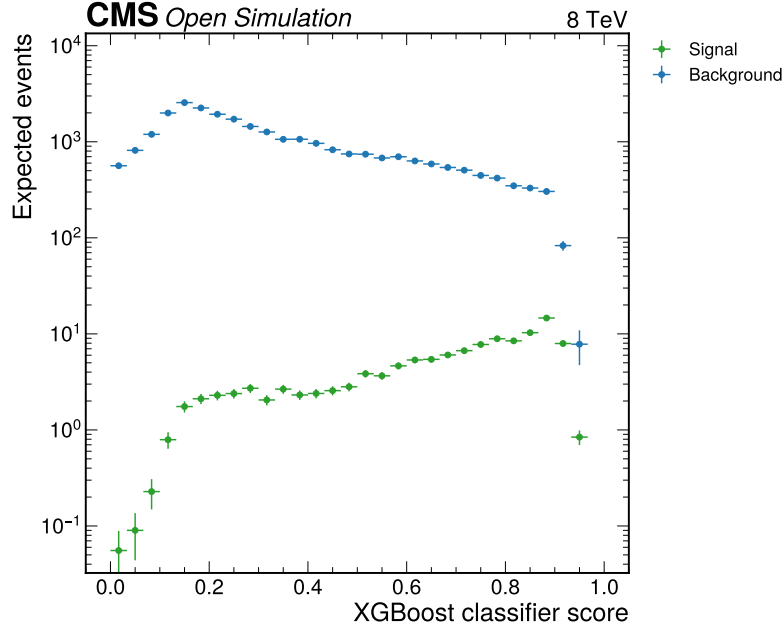


Figure 12: Expected classifier score templates. The figure shows the trained classifier score response for signal and background before observed-data fitting. It checks that the retained classifier has the intended signal/background ordering.

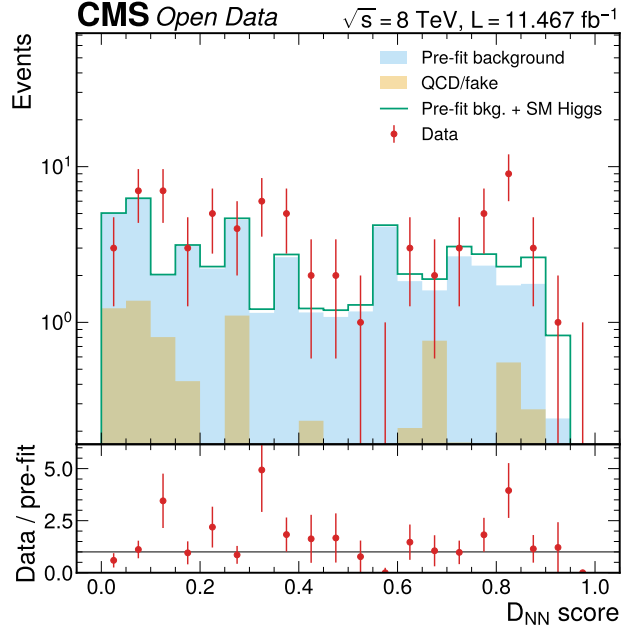


Figure 13: D_{NN} score pre-fit validation in the VBF category. The plot compares data with the nominal classifier-score background plus Standard Model Higgs prediction before profiling nuisance parameters in the secondary score fit. The lower panel is Data/pre-fit.

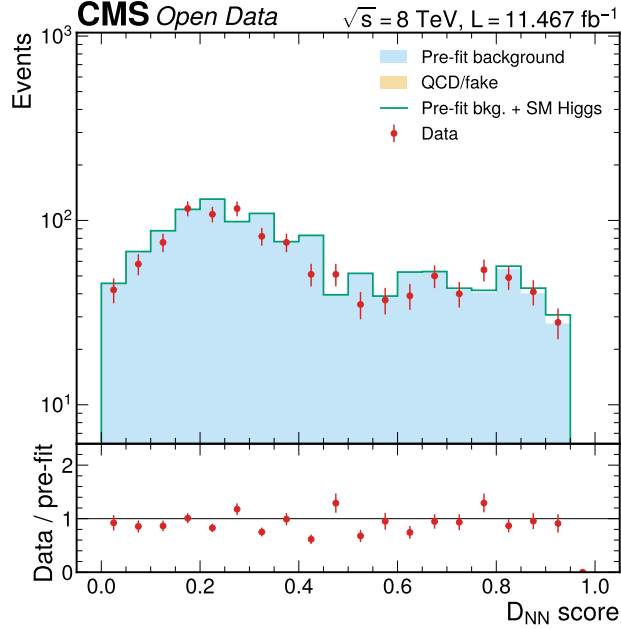


Figure 14: D_{NN} score pre-fit validation in the boosted category. The plot shows the boosted-category nominal classifier-score prediction before the post-fit profile update. It uses the same score binning as the secondary workspace.

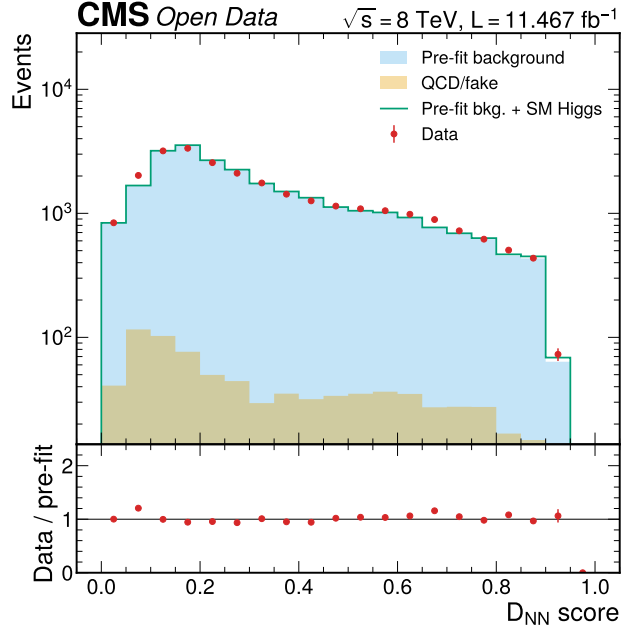


Figure 15: D_{NN} score pre-fit validation in the zero-jet category. The plot shows the statistically dominant classifier-score category before nuisance profiling. The lower panel is Data/pre-fit.

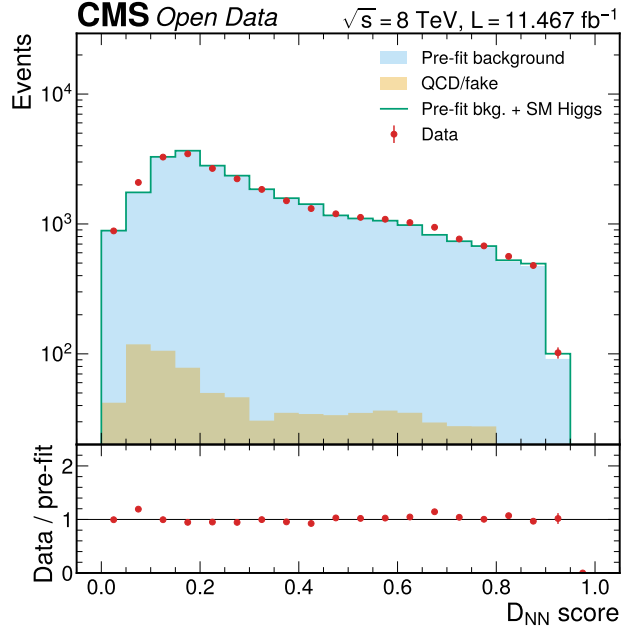


Figure 16: D_{NN} score combined pre-fit validation. The plot sums the VBF, boosted, and zero-jet nominal classifier-score predictions and compares them with the summed observed data before profiling. The lower panel is the combined Data/pre-fit ratio.

5 Background Model

The W+jets normalization is constrained from a high- m_T control region. With the current inputs the full-data scale factor is 0.9190 ± 0.0139 . The VBF background scale from the VBF-like non-signal control region is 0.3448 ± 0.0612 . The reducible QCD/fake component is estimated from same-sign data after subtracting non-QCD simulation and transferring into the signal region.

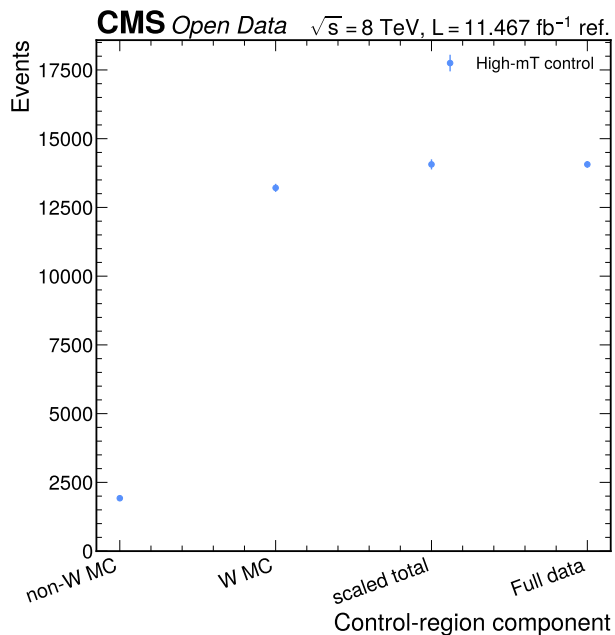


Figure 17: W high- m_T control-region scale. The figure shows the non-W MC, nominal W MC, scaled control-region prediction, and observed data in the high- m_T control region. This scale is propagated to both retained workspaces.

6 Systematic Uncertainties

The retained nuisance program covers luminosity, tau/open-data acceptance, DY/Z normalization, W+jets control normalization, VBF background control normalization, same-sign QCD/fake transfer, and MC statistical uncertainties. The exact current values are read from the Phase-4c JSON artifact:

Source	Current value
dy_ztautau_open_data	30% in boosted/zero-jet and 50% in VBF
lumi_2012	2.6%
qcd_ss_transfer_nn	1.07652 relative from same-sign/low-sideband data
qcd_ss_transfer_visible	0.315486 relative from same-sign/low-sideband data
tau_open_data_acceptance	15%
vbf_background_control	0.177463 relative from VBF-like top-btag non-signal CR
wjets_high_mt_control	0.015166 relative from full high-mT data CR

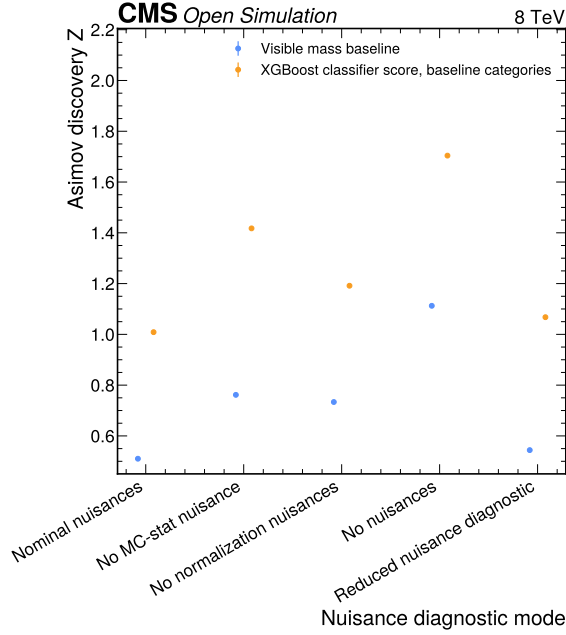


Figure 18: Sensitivity nuisance audit. The figure summarizes the expected sensitivity response to the retained nuisance model. It is used to confirm that the current results are not driven by a hidden no-systematics configuration.

7 Statistical Model

Both retained results use binned HistFactory-style likelihoods implemented with pyhf (Heinrich et al. 2021). For category c and bin b , the expected count is

$$\nu_{cb}(\mu, \theta) = \mu s_{cb}(\theta) + \sum_k b_{kcb}(\theta), \quad (2)$$

where s_{cb} is the sum of ggH and VBF signal templates and b_{kcb} are the background templates. The likelihood is the product of Poisson terms for the binned data and Gaussian/log-normal constraints for nuisance parameters. Upper limits use the modified frequentist CLs construction (Read 2002) with asymptotic formulae (Cowan et al. 2011).

The expected median limit from the earlier expected-only reference point is 2.90409966080617. The current observed limits are computed from the full data with the updated local selection and templates.

8 Full-Data Results

Category	Data	Background	Signal	Data/background	Chi2/ndf
vbf	70	53.37	4.13	1.312	1.999
boosted	1138	1249.00	11.28	0.911	0.492
zero_jet	25920	26732.68	106.26	0.970	0.126

The combined visible-mass validation gives data/background 0.968 and chi2/ndf 0.873. The combined D_{NN} validation gives data/background 1.003 and chi2/ndf 0.685.

The visible-mass baseline goodness-of-fit is checked with fitted toys from the post-fit pyhf/cabintetry model. The observed fit uses `cabintetry.fit.fit(..., goodness_of_fit=True)`, and each of the 100 toys is refit with cabintetry before evaluating the saturated-model GoF statistic. The observed statistic is 24.7215, the cabintetry saturated GoF p-value is 0.1011, and the toy p-value is 0.1683. The toy central 95% interval is 8.2208 to 31.6523, so the observed GoF is **inside** the toy distribution and the check is **PASS**.

GoF quantity	Value
Number of fitted toys	100
Toy fit failures	0
Observed saturated GoF statistic	24.7215
Cabinetry saturated GoF p-value	0.1011
Toy median statistic	16.3962
Toy central 95% interval	8.2208 to 31.6523
Fitted-toy p-value	0.1683
Decision	PASS

The visible-mass baseline fit gives

$$\hat{\mu}(m_{vis}) = 2.4740, \quad \mu < 5.9262 \text{ (95\% CLs)}, \quad (3)$$

with median expected limit 3.6919. The D_{NN} result gives

$$\hat{\mu}(D_{NN}) = 1.6160, \quad \mu < 3.5769 \text{ (95\% CLs)}, \quad (4)$$

with median expected limit 1.8069.

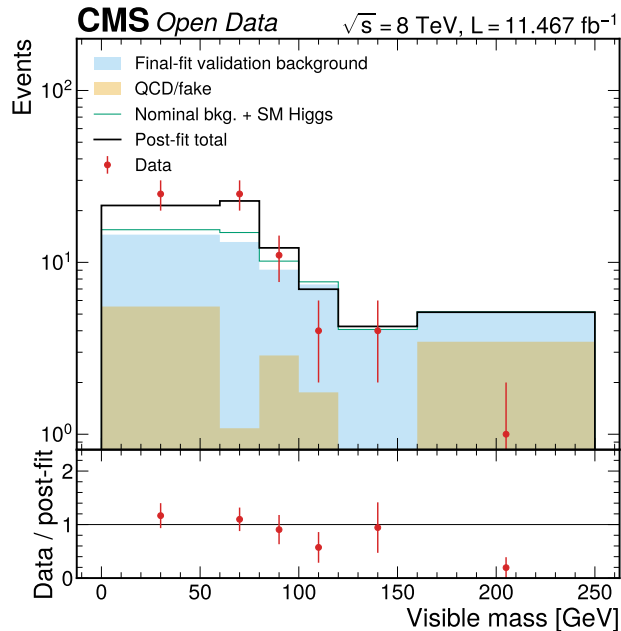


Figure 19: Visible-mass baseline post-fit validation in the VBF category. The plot compares data with the post-fit visible-mass total from the retained baseline workspace, with the nominal background plus Standard Model Higgs template overlaid for reference. The lower panel shows the Data/post-fit ratio.

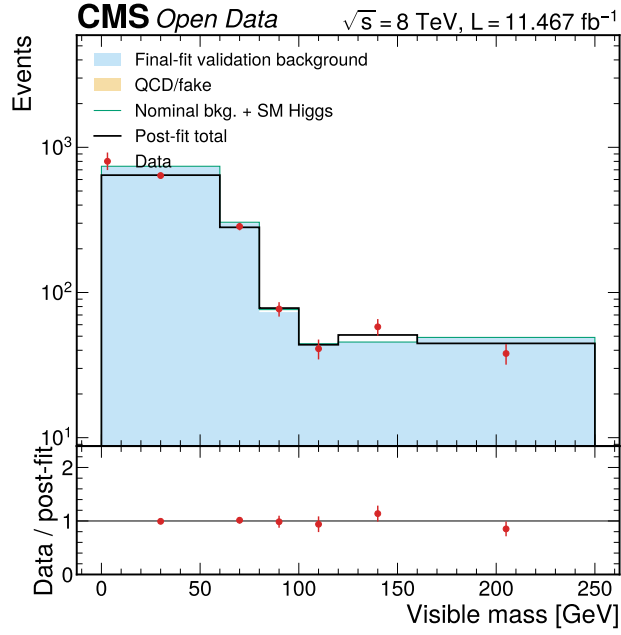


Figure 20: Visible-mass baseline post-fit validation in the boosted category. The plot compares data with the post-fit visible-mass total from the retained baseline workspace. This category has substantially more events than the VBF category and constrains the broad visible-mass shape.

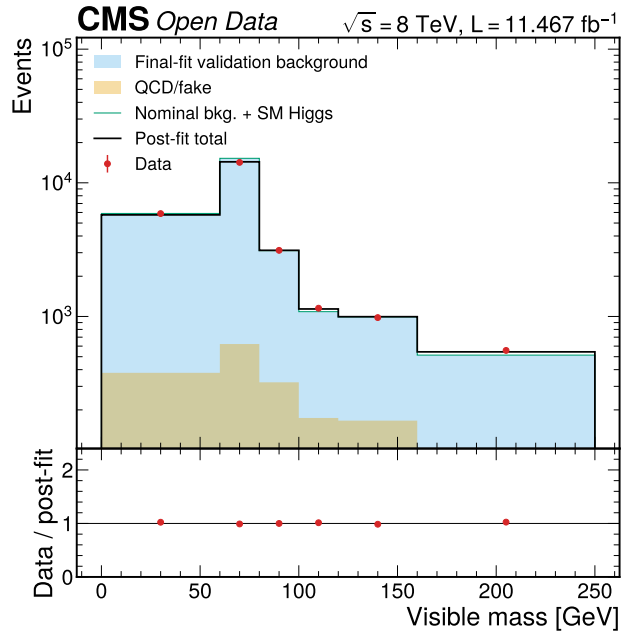


Figure 21: Visible-mass baseline post-fit validation in the zero-jet category. The plot compares data with the post-fit visible-mass total from the retained baseline workspace in the statistically dominant category. This category controls the global normalization behavior of the baseline workspace.

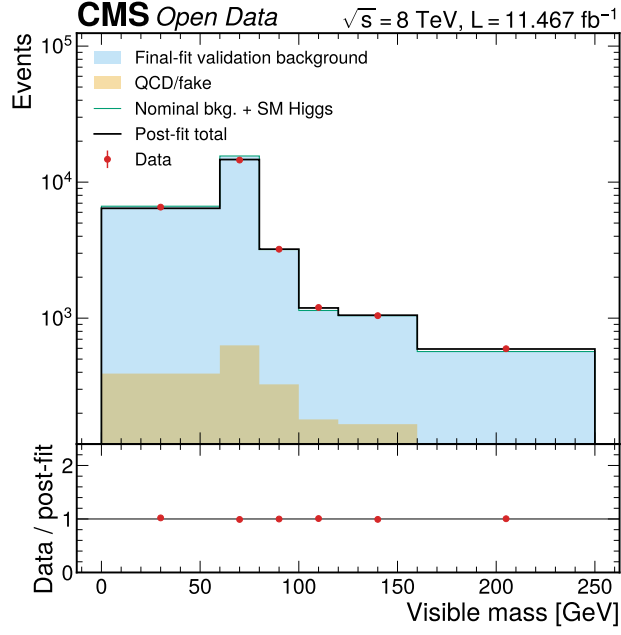


Figure 22: Combined visible-mass baseline post-fit validation. The plot sums the VBF, boosted, and zero-jet post-fit visible-mass model expectations and compares them with the summed observed data. The lower panel is the combined Data/post-fit ratio used as the global baseline Data/MC check.

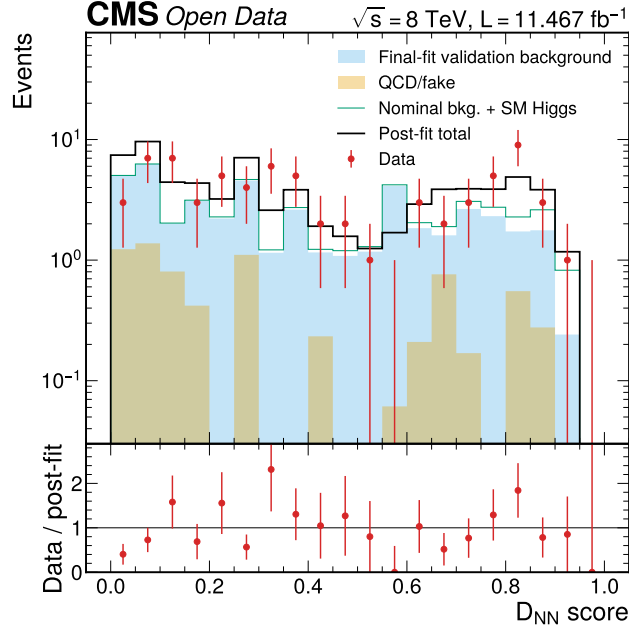


Figure 23: D_{NN} score post-fit validation in the VBF category. The plot compares data with the post-fit classifier-score total in the VBF category. The same score definition and binning are used in the other categories.

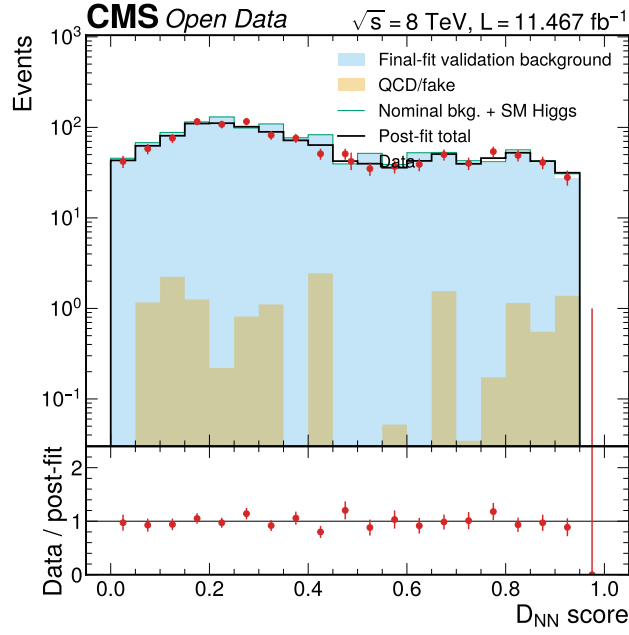


Figure 24: D_{NN} score post-fit validation in the boosted category. The plot compares data with the post-fit classifier-score total in the boosted category. The ratio panel checks whether the classifier ordering is modelled in observed data.

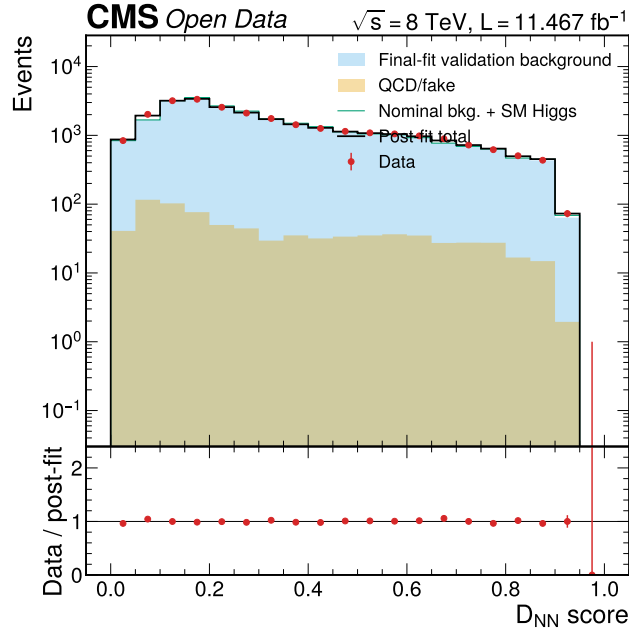


Figure 25: D_{NN} score post-fit validation in the zero-jet category. The plot compares data with the post-fit classifier-score total in the zero-jet category. The category has the largest statistical weight in the combined classifier fit.

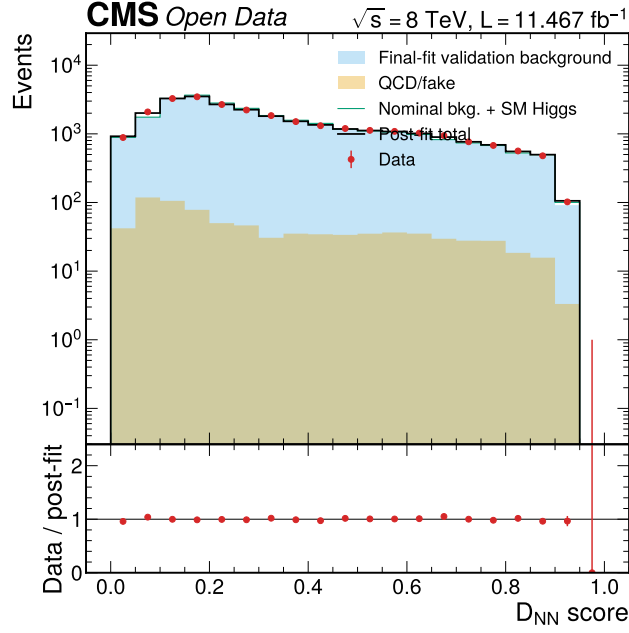


Figure 26: Combined D_{NN} score post-fit validation. The plot sums the VBF, boosted, and zero-jet post-fit classifier-score expectations and compares them with the summed observed score distribution. The lower panel is the combined Data/post-fit ratio used as the global score Data/MC check.

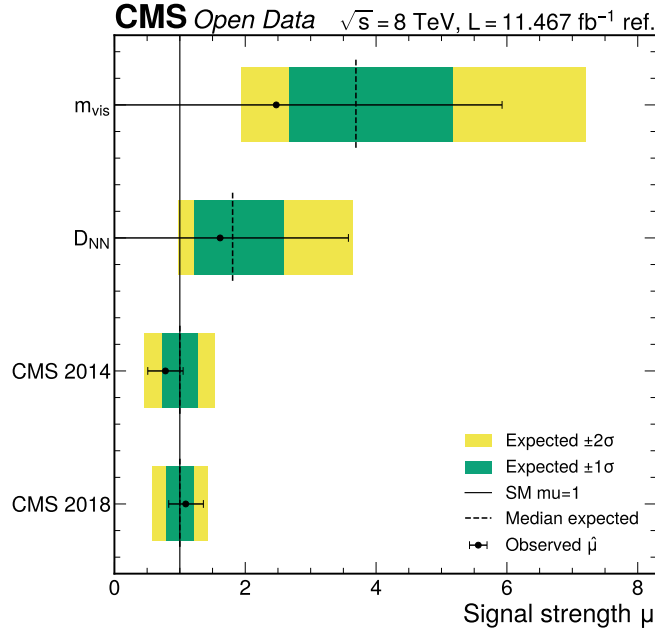


Figure 27: Observed result summary. The figure compares the visible-mass baseline, retained D_{NN} result, CMS 2014 result, and CMS 2018 result on the same signal-strength scale. Each row uses the same convention: black observed point with horizontal uncertainty, green and yellow expected one- and two-standard deviation bands, a black dashed median-expected marker, and the common Standard Model line at $\mu = 1$.

9 Comparison To Published Measurements

The CMS Run-1 H to tau tau analysis and later CMS combination use more final states, more luminosity, embedded backgrounds, and a much richer calibration program than this single-channel public-data analysis (CMS Collaboration 2014, 2018). The correct comparison is therefore qualitative: the present analysis demonstrates a reproducible template-fit chain and produces limits that are much weaker than the published measurements. The retained D_{NN}

result is the more sensitive of the two current outputs by expected median limit, while the visible-mass baseline remains the cut-based reference.

10 Validation And Limitations

The current run validates that both pyhf workspaces construct, that both fits evaluate, that the D_{NN} template uses `mva_score_xgboost`, and that the classifier score uses 20 uniform bins. Remaining limitations are inherited from the reduced public-input setting: the absence of embedded $Z \rightarrow \tau\tau$, incomplete rare backgrounds, simplified object systematics, and reliance on control-derived nuisance parameters.

11 Reproduction Contract

The current machine-readable result artifacts are:

Artifact	Purpose
<code>phase4_inference/4c_observed/outputs/baseline_visible_result.json</code>	baseline result
<code>phase4_inference/4c_observed/outputs/nn_score_result.json</code>	obtained D_{NN} result
<code>phase4_inference/4c_observed/outputs/pyhf_workspace_baseline_visible.json</code>	baseline visible workspace
<code>phase4_inference/4c_observed/outputs/pyhf_workspace_nn_score.json</code>	nn score workspace
<code>phase4_inference/4c_observed/outputs/visible_observed_templates.json</code>	visible observed templates
<code>phase4_inference/4c_observed/outputs/nn_score_observed_templates.json</code>	nn score observed templates

The focused rebuild commands are `pixi run phase3-select`, `pixi run phase3-mva`, `pixi run phase3-sensitivity`, `pixi run phase3-plots`, `pixi run phase4c-all`, and `pixi run phase5-docs`.

- CMS Collaboration. 2014. “Evidence for the 125 GeV Higgs Boson Decaying to a Pair of Tau Leptons.” *Journal of High Energy Physics* 2014 (5): 104. [https://doi.org/10.1007/JHEP05\(2014\)104](https://doi.org/10.1007/JHEP05(2014)104).
- CMS Collaboration. 2018. “Observation of the Higgs Boson Decay to a Pair of Tau Leptons with the CMS Detector.” *Physics Letters B* 779: 283–316. <https://doi.org/10.1016/j.physletb.2018.02.004>.
- Cowan, Glen, Kyle Cranmer, Eilam Gross, and Ofer Vitells. 2011. “Asymptotic Formulae for Likelihood-Based Tests of New Physics.” *European Physical Journal C* 71: 1554. <https://doi.org/10.1140/epjc/s10052-011-1554-0>.
- Heinrich, Lukas, Matthew Feickert, Giordon Stark, and Kyle Cranmer. 2021. “Pyhf: Pure-Python Implementation of HistFactory Statistical Models.” *Journal of Open Source Software* 6 (58): 2823. <https://doi.org/10.21105/joss.02823>.
- Read, A. L. 2002. “Presentation of Search Results: The CLs Technique.” *Journal of Physics G* 28: 2693–704. <https://doi.org/10.1088/0954-3899/28/10/313>.